

SYED MUHAMMAD ABUBAKAR

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Professional Summary

Machine Learning Engineer with production experience deploying generative AI, LLM, and computer vision systems, combining data science, model development, and MLOps. Backed by an Electrical Engineering degree and an Erasmus Mundus Master's in Sustainable Systems Engineering. Track record spans applied research (MICCAI, ICES JMS submissions), LLM fine-tuning and RAG, MLOps-driven deployment, and hands-on exposure to large-scale compute infrastructure at Google. Seeking ML/AI engineering roles building intelligent systems end to end.

Work Experience

Data Center Technician Intern

June 2026 - Sep 2026

Google, Belgium

- Completed rotations across five operational domains (Deployments, Network Delivery, Interrupts, Networking, Data Security), building ground-level understanding of the infrastructure powering large-scale ML workloads
- Diagnosed and repaired production server hardware - swapping failed DIMMs, BIOS chips, and QSFP optics - and resolved backbone fiber link faults using OTDR diagnostics to restore capacity
- Executed secure media sanitization of 300+ storage devices and verified live traffic flows before hardware decommissioning, ensuring zero customer impact during infrastructure changes

Machine Learning Engineer

Aug 2024 - July 2025

Dcube Technologies, Islamabad, Pakistan

- Developed a generative AI audio enhancement system for broadcast audio from about 10 official NFL game recordings, achieving 96% recall in audio event detection, using GANs for data augmentation and a real-time parameter control interface for studio engineers
- Built LLM-based modules for *Fabric*, an e-commerce platform, enabling natural-language querying and automated business-intelligence summaries from sales data, fine-tuned to each client business's specific workflows
- Researched agentic AI architectures for autonomous fish detection and classification, building a multi-agent vision pipeline for underwater species monitoring and adaptive task allocation
- Led *Agentic Active Learning* (MICCAI 2025 submission): a Qwen2-VL agent describes 705 dental X-rays, ClinicalBERT embeds the text, and a multimodal CoreSet query fuses it with YOLOv11 image features, matching full-data detection of 4 dental diseases with 20-30% less labeled data on DENTEX (dentists corrected only 9% of agent outputs)
- Spearheaded containerized production deployments using Docker and Kubernetes, serving models through FastAPI endpoints, including an attendance-tracking system in daily use for 100+ students, ensuring scalable, fault-tolerant operation
- Presented real-time AI audio proof-of-concept demonstrations to FOX Sports and global stakeholders
- Improved team productivity by establishing modular Git-based codebases, MLflow and Weights & Biases experiment tracking, and CI/CD pipelines

METI AI/IT Intern

Nov 2024 - Dec 2024

ARS Corporation, Tokyo, Japan

- Developed and deployed serverless applications with AWS SAM for real-time remote monitoring, streamlining deployment and reducing system complexity
- Implemented secure remote networking with ZeroTier, achieving 99.9% uptime for remote device access and management
- Containerized embedded configurations with Docker, enhancing portability and streamlining deployment
- Optimized real-time data processing pipelines on Raspberry Pi and Armadillo G3L edge devices, bridging cloud services with constrained hardware

Education

European Master in Sustainable Systems Engineering (Erasmus Mundus)

Aug 2025 - Aug 2027 (expected)

Université de Technologie de Compiègne (France); University of Genoa (Italy)

- Systems Engineering track (SoSE); mobilities completed in France (GPA 3.70/4.00) and Albania (GPA 9.41/10); final mobility in Computer Engineering at the University of Genoa (2026/27)
- Selected coursework: Industrial Automation, Model-Based Systems Engineering (SysML), Scientific Machine Learning

Bachelor of Science in Electrical Engineering

Sep 2020 - June 2024

National University of Sciences and Technology, Islamabad, Pakistan

- Graduated with Distinction (CGPA: 3.62/4.00); recipient of the FAST Cables Ltd. Sustainability Award
- Thesis: Fish Biomass Estimation in Underwater Unconstrained Environments
- Relevant coursework: Machine Learning, Signal Processing, Computer Networks, Digital Logic Design, Control Systems

Research Experience

Undergraduate Research Assistant (Fish Biomass Estimation, Final Year Project)

June 2023 - June 2024

Optimization and Machine Learning (OPTIMAL) Lab, NUST, Islamabad, Pakistan | [GitHub](#)

- Collaborated with the Machine Vision Group, University of Western Australia, on deep learning-based fish biomass estimation in unconstrained underwater environments, funded by an Australian Government research grant
- Fused YOLOv8 with a Gaussian-mixture foreground change estimator (FCE), reaching 97.5% detection F1 and 96.3% species accuracy across 15 species on LifeCLEF 2015; tracked fish with Hungarian-algorithm assignment
- Developed an attention U-Net for monocular depth estimation that achieved lower RMSE than prior methods, trained on 1.5M+ KITTI/NYU images and transfer-learned to underwater video
- Estimated biomass from depth-scaled fish length and species length-weight regression (5.67% error vs. reference sizes); co-authored the *DeepDive* manuscript submitted to ICES JMS (ICESJMS-2024-271)

Undergraduate Research Intern

June 2021 - Sep 2022

TUKL-NUST R&D Center, Islamabad, Pakistan

- Selected through a competitive programming test; specialized in deep learning for signal processing and computer vision
- Implemented the Splerge algorithm for table structure analysis, combining split-and-merge techniques with Scale Factorization Convolutional, Residual Prediction, and Coarse Prediction Networks to reach 95.9% precision in document table extraction
- Designed an AlexNet+MLP model for EEG anomaly detection, achieving 89% accuracy in identifying irregular brain signals
- Built and deployed a real-time face recognition entry system (OpenCV, TensorFlow) adopted for daily access control by roughly 30 lab members at the TUKL Lab
- Ran distributed training experiments on remote Ubuntu GPU servers and built annotated datasets with LabelMe and Roboflow

Projects

3D Object Pose Estimation with Embedded AI Processing | [GitHub](#)

Sep 2025 - Jan 2026

- Built 6DOF pose estimation for conveyor-belt objects from 15 YDLidar G2 point-cloud slices (150 statistical features)
- Trained an MLP regressor with 5-fold cross-validation: 2.0 mm position and 0.68° rotation MAE on held-out test data
- Deployed on Jetson Orin Nano as a ROS2 pipeline with ONNX Runtime inference for real-time pose output

ChatWithPDFs: Conversational AI for PDF Interaction | [GitHub](#)

Aug 2024

- Built a RAG-based question-answering tool over PDF documents using LangChain and the Gemini API
- Engineered the retrieval stack with GoogleGenerativeAI embeddings and FAISS vector storage, deployed via Streamlit with a custom QA chain

EmoSenseVision: Real-Time Facial Emotion Recognition | [GitHub](#)

Nov 2023 - Dec 2023

- Developed a custom VGG-inspired CNN for facial expression analysis reaching 68% accuracy with regularization tuning, deployed for live webcam inference with Haar cascade face detection restricting predictions to detected faces

Serverless Text-to-Audio Conversion Pipeline

Nov 2024

- Designed an asynchronous serverless architecture on AWS Lambda and Amazon Polly converting text posts to natural-sounding audio, with secure storage and user-accessible delivery via S3
- Orchestrated the workflow across multiple Lambda functions with unique ID generation in DynamoDB, decoupling text ingestion from speech synthesis for fault-tolerant processing

Skills

Programming: Python, C/C++, MATLAB, Bash, SQL, Git; API Development (REST, FastAPI), Serverless Applications

Deep Learning: PyTorch, TensorFlow, Keras, Scikit-Learn, OpenCV, Hugging Face Transformers

Data Science: Pandas, NumPy, Jupyter, Exploratory Data Analysis, Feature Engineering, Statistical Modeling, Model Evaluation & Monitoring

GenAI & LLMs: LLM Fine-Tuning (LoRA, PEFT), Vision-Language Models, LangChain, RAG, FAISS, Vector Embeddings, Agentic Pipelines, Prompt Engineering, Gemini/OpenAI APIs

MLOps & Cloud: Docker, Kubernetes, CI/CD, MLflow, Weights & Biases, ONNX, TensorRT, AWS (Lambda, S3, DynamoDB, SAM), Roboflow, LabelMe

Edge & Systems: Jetson Orin Nano, Raspberry Pi, ROS2, IoT, ZeroTier, Real-Time Inference, Data Center Operations, Linux/Ubuntu

Honors / Awards

- **Erasmus Mundus Scholarship**, European Commission - fully-funded Joint Master's Scholarship for EMSSE (Apr 2025)
- **Fully-funded METI AI/IT Japan Internship** - selected as the only Pakistani from a global pool of 15,000 candidates (Nov 2024)
- **FAST Sustainability Award** (PKR 200,000) - most sustainable undergraduate Final Year Project (Oct 2024)
- **Australian Government Research Grant** - fish biomass estimation with UWA (Mar 2023)
- **High Achievers Award - Gold Medal** (2020)
- **Distinction in O-Level Computer Science** - among top performers in Northern Pakistan (2018)